



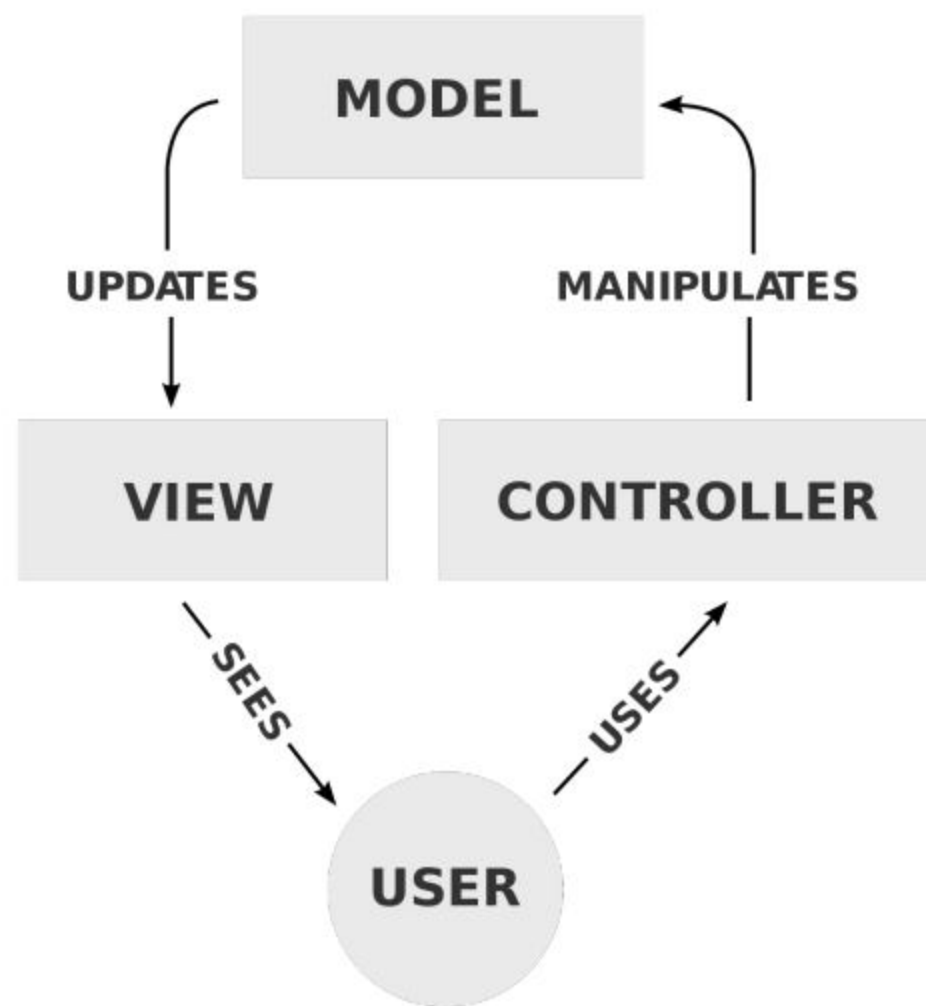
Week 5 Lecture 1

▼ Class	BSCCS2003
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# Module #	27
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☰ Week #	5

MVC Origins

Model-View-Controller

- Design pattern: or collection of design patterns
- Originally introduced in the context of GUI design in Smalltalk-80
- Many different variants, interpretations, ...



By RegisFrey - Own work, Public Domain, Wikipedia

From: Trygve Reenskaug

Date: 10 December 1979

MODELS-VIEWS-CONTROLLERS

MODELS

Models represent knowledge. A model could be a single object (rather uninteresting), or it could be some structure of objects. ~~The proposed implementation supports knowledge represented in something resembling semantic nets (If I understand Laura correctly)~~

VIEWS

A view is a (visual) representation of its model. It would ordinarily highlight certain attributes of the model and suppress others. It is thus acting as a *presentation filter*.

A view is attached to its model (or model part) and gets the data necessary for the presentation from the model by asking questions. It may also update the model by sending appropriate messages. All these questions and messages have to be in the terminology of the model, the view will therefore have to know the semantics of the attributes of the model it represents. (It may, for example, ask for the model's identifier and expect an instance of Text, it may not assume that the model is of class Text)

CONTROLLERS

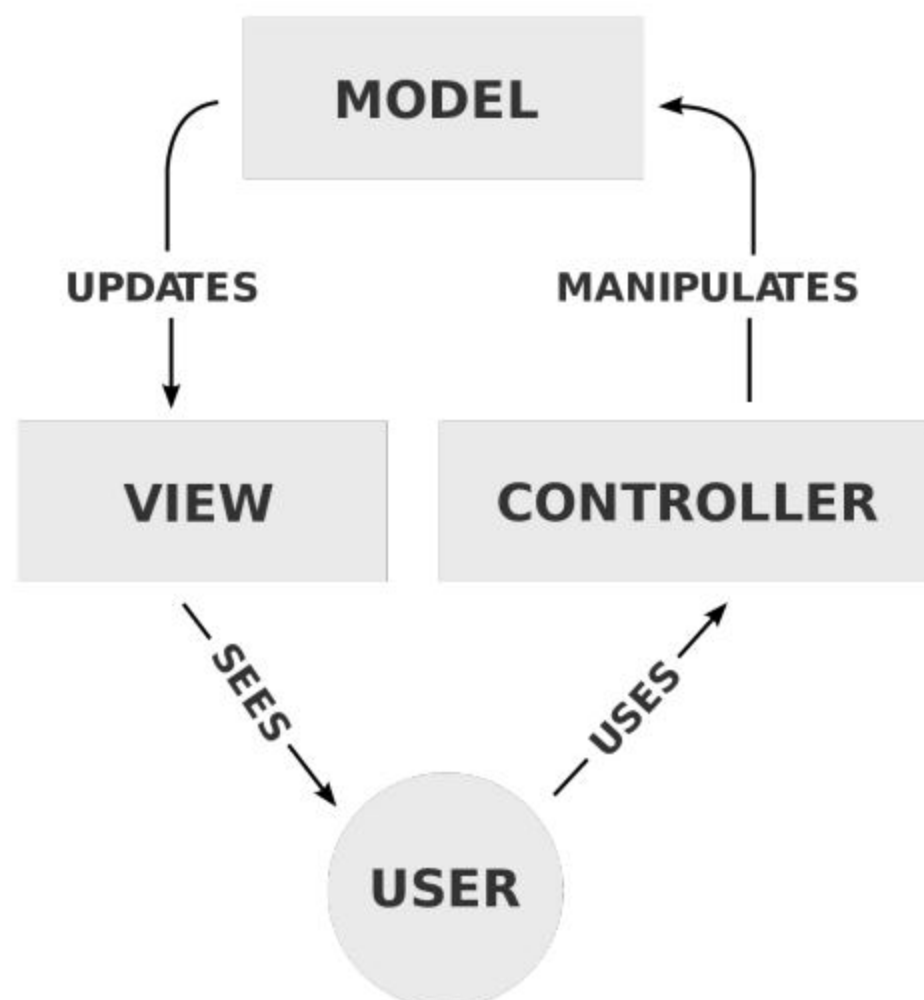
A controller is the link between a user and the system. It provides the user with input by arranging for relevant views to present themselves in appropriate places on the screen. It provides means for user output by presenting the user with menus or other means of giving commands and data. The controller receives such user output, translates it into the appropriate messages and pass these messages on .to one or more of the views.

A controller should never supplement the views, it should for example never connect the views of nodes by drawing arrows between them.

Conversely, a view should never know about user input, such as mouse operations and keystrokes. It should always be possible to write a method in a controller that sends messages to views which exactly reproduce any sequence of user commands.

General Concept - Action

- Take action in response to user input
- Communicate with the model, extract the view



Applicability

- Originally designed for GUI applications
- Separation of concerns - model vs views - and connection through controller
- State of interaction maintained as part of overall system memory

Web?

- Server does not maintain the state of the client
- Client is pure front-end to the user
- Some of the analogies break down - hence many variants of MVC

Present Context

- MVC is a good conceptual framework to understand separation of concerns
- Breaks down if applied too rigidly
- The web in general does not have the close knit structure of GUI applications needed for MVC
- Other aspects of static typing and type inference of objects also broken in Python-like languages

Apply basic learnings from MVC, but be prepared to stretch